

Andrea López López

PhD Candidate, Aerospace Engineering Sciences
Ann and H.J. Smead Department of Aerospace Engineering Sciences
University of Colorado Boulder

andrea.lopez@colorado.edu | Google Scholar | LinkedIn | Website

Education

Ph.D., Aerospace Engineering Sciences	Aug. 2021 – expected Spring 2027
University of Colorado Boulder	
Advisors: Prof. Hanspeter Schaub, Prof. Jay W. McMahon	
Dissertation title: <i>Observability-Informed Adaptive Gaussian Mixture Filtering for Cislunar Angles-Only Relative Navigation</i>	
Comprehensive exam passed March 2026.	
Graduate Certificate in Space Domain Awareness	
M.S., Aerospace Engineering Sciences , University of Colorado Boulder	2024
M.S., Aerospace Engineering , Université de Liège, Belgium	2019
M.S., Aeronautical Engineering , Universidad Politécnica de Madrid, Spain	2019
B.S., Aerospace Engineering , Universidad Politécnica de Madrid, Spain	2016

Core Skills

Core Competencies: nonlinear state estimation, orbit determination, optical and angles-only navigation, rendezvous and proximity operations, formation flying, guidance, navigation, and control (GNC), trajectory design, space domain awareness, uncertainty quantification, systems engineering.

Software: Python, MATLAB, Rust (familiarity), Git, unit testing, continuous integration.

Languages: Galician and Spanish (native), English (professional), French (B2/C1).

Research and Professional Experience

Graduate Research Assistant Aug. 2021 – present
Autonomous Vehicle Systems (AVS) Laboratory and Orbital Research Cluster for Celestial Applications (ORCCA) Laboratory, University of Colorado Boulder

- Flight dynamics affiliate at Laboratory for Atmospheric and Space Physics (LASP): Contributor to Scarabaeus, an open-source orbit determination and navigation tool (Python front end, Rust backend) developed by ORCCA to support the Emirates Mission to the Asteroid Belt (EMA).
- Cislunar angles-only relative navigation: nonlinear observability analysis, Gaussian mixture filtering (GMSRUKF), CR3BP relative dynamics.
- Phase I STTR with In Orbit Aerospace: investigated electroadhesion for spacecraft docking and debris capture.
- AFOSR-funded research on touchless electrostatic potential sensing of spacecraft using X-ray and electron-based methods; wide field-of-view binary X-ray sensing for space object detection.

Young Graduate Trainee, Quality, Dependability & Safety Jun. 2019 – Jun. 2021
European Space Agency (ESA-ESTEC), Noordwijk, Netherlands

- Performed technical risk assessments in the Concurrent Design Facility; improved risk management methods and tooling (VBA, SQL).
- Supported RAMS analyses and verification activities across ESA projects (PDR, CDR, QR, AR).

Publications

Selected; full list available on website.

Journal Articles

- **A. López**, J. Kulik, J. W. McMahon, “Observability Analysis of Cislunar Angles-Only Navigation: Metrics, Geometry, and Filter Implications,” *Journal of the Astronautical Sciences*, 2026. (In Preparation)
- **A. López**, J. W. McMahon, H. Schaub, “Square Root Filters for Cislunar Angles-Only Relative Orbit Estimation,” *Journal of the Astronautical Sciences*, 2026. doi:10.1007/s40295-026-00617-8
- **A. López**, H. Schaub, “Detecting Space Objects with Binary Wide Field-of-View X-Ray Sensing,” *Journal of Spacecraft and Rockets*, 2025. (In Review)
- J. Hammerl, **A. López**, Á. Romero Calvo, H. Schaub, “Touchless Potential Sensing of Differentially-Charged Spacecraft Using X-Rays,” *Journal of Spacecraft and Rockets*, 2023. doi:10.2514/1.A35492

First Author Conference Papers

- **A. López**, J. Kulik, J. W. McMahon, “Observability Considerations for Cislunar Angles-Only Relative Navigation,” AMOS Conference, Maui, HI, Sep. 2026.
- **A. López**, J. W. McMahon, H. Schaub, “Gaussian Mixture Square Root Filters for Cislunar Angles-Only Relative Orbit Determination,” AAS/AIAA Astrodynamics Specialist Conference, Boston, MA, Aug. 2026.
- **A. López**, H. Schaub, “Touchless Time-Varying Electrostatic Potential Sensing in Orbit: Fusion of X-ray and Electron Methods,” Spacecraft Charging and Technology Conference, Avignon, France, May 2024.
- **A. López**, H. Schaub, “Relative Orbit Estimation with Wide Field of View Binary X-ray Sensing,” AMOS Conference, Maui, HI, Sep. 2023.
- **A. López**, J. Hammerl, H. Schaub, “Dynamic Detection of Nearby Space Objects with Binary Wide Field of View X-Ray Sensing,” AIAA SciTech Forum, National Harbor, MD, Jan. 2023. doi:10.2514/6.2023-1397
- **A. López**, J. Hammerl, H. Schaub, “Detecting Space Objects With Binary Wide Field of View X-Ray Sensing,” AAS/AIAA Astrodynamics Specialist Conference, Charlotte, NC, Aug. 2022.

Awards and Fellowships

NASA FINESST Fellowship (Grant No. 80NSSC23K1623) 2023 – 2026

Amelia Earhart Fellowship, Zonta International 2024

Service and Leadership

Member/President, Aerospace Graduate Student Organization (AGSO) 2021 – 2025
Smead Aerospace, University of Colorado Boulder. President 2023 – 2025; previously co-chair and first-year representative. Ran professional development, mentoring, and outreach programming; represented graduate students in departmental decision-making.

Member, Smead Aerospace Graduate Committee 2022 – 2025
Graduate student representative on the departmental committee for ~3 years.

Co-founder, CCAR Student Seminar Series 2022 – present
Colorado Center for Astrodynamics Research, University of Colorado Boulder. Co-founder and organizer of student seminar series, hosting 4–5 seminars per semester.